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Developing a Text-to-Speech Model for the Algerian Dialect

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Dedication

Alasmer Anas's Dedication

Dedicated with love and gratitude to:

My beloved parents, Ramez and Rana,

My beloved grandmother Ghadah,

*My dear and inspiring big brother Sami and my little brother
Ghaith,*

My aunts, uncles, and all my cousins from both sides of the family,

*And my close friends: Mohamed, Ziad, Chahine, Faycal, Raouf,
Fares, Adel and Chems Eddine.*

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I am deeply grateful to my family for their belief, support, and guidance.

I want to express my gratitude to my friend Benmati Ziad for his help, support, and encouragement throughout this journey.

Abstract

Algerian Arabic, an under-resourced and linguistically distinct dialect, poses significant challenges for current Text-to-Speech (TTS) systems due to limited data and lack of specialized models. This final project tackles this gap by fine-tuning **Coqui XTTS**, a *state-of-the-art*, cross-lingual TTS system renowned for voice cloning and expressivity. To the best of our knowledge, no existing model addresses Algerian Arabic speech synthesis, making this work a novel contribution.

The approach involved building a custom dataset of Algerian Arabic recordings and corresponding transcriptions. Transfer learning was applied to adapt the pre-trained XTTS architecture to this dialect. The objective is to produce high-quality, natural, and contextually relevant speech in Algerian Arabic, while preserving vocal characteristics.

Evaluation will focus on the intelligibility, naturalness, and expressiveness of the synthesized speech. This project demonstrates the potential of advanced multilingual TTS models like XTTS to serve low-resource languages and dialects, paving the way for more inclusive speech technologies.

Keywords: Algerian Arabic, XTTS, State-of-the-Art, Speech Synthesis, Transfer Learning, Low-Resource

Résumé

L'arabe algérien, dialecte sous-doté en ressources et linguistiquement spécifique, représente un défi important pour les systèmes actuels de synthèse vocale (TTS) en raison du manque de données et de modèles spécialisés. Ce projet de fin d'études comble cette lacune en affinant **Coqui XTTS**, un système TTS multilingue *à la pointe de la technologie*, réputé pour le clonage vocal et son expressivité. À notre connaissance, aucun modèle existant ne cible la synthèse vocale en arabe algérien, ce qui rend ce travail inédit.

La méthode a consisté à créer un corpus personnalisé de paroles et de transcriptions en arabe algérien. L'apprentissage par transfert a été utilisé pour adapter l'architecture pré-entraînée XTTS à ce dialecte. L'objectif est de générer une parole naturelle, fluide et contextuellement adaptée, tout en conservant les caractéristiques vocales.

L'évaluation portera sur l'intelligibilité, la naturalité et l'expressivité de la synthèse vocale. Ce projet illustre le potentiel des modèles TTS multilingues avancés comme XTTS pour les langues et dialectes à faibles ressources, ouvrant la voie à des technologies vocales plus inclusives.

Mots-clés : Arabe algérien, XTTS, Technologie de pointe, Synthèse vocale, Apprentissage par transfert, Faibles ressources

ملخص

اللهجة الجزائرية، وهي لهجة محدودة الموارد وذات خصوصية لغوية، تمثل تحدياً كبيراً لأنظمة تحويل النص إلى كلام (TTS) الحالية بسبب قلة البيانات وغياب النماذج المتخصصة. يعالج هذا المشروع هذا النقص من خلال الضبط الدقيق لنموذج **Coqui XTTS**، وهو نظام *TTS* متعدد اللغات وواحد من أحدث النماذج، معروف بقدرته على استنساخ الصوت والتعبير الصوتي. ولأول مرة، يتم تكييف هذا النموذج مع اللهجة الجزائرية، مما يجعل هذا العمل جديداً ومميزاً. تضمن المنهجية إنشاء مجموعة بيانات مخصصة من التسجيلات الصوتية والنصوص باللهجة الجزائرية، ثم تطبيق تقنيات التعلم بالنقل على بنية *XTTS* المسبقة التدريب. الهدف هو إنتاج كلام عالي الجودة، طبيعي، ومتوافق مع السياق، مع الحفاظ على الخصائص الصوتية المميزة للهجة.

سيتم تقييم نجاح هذا الضبط الدقيق بناءً على وضوح الكلام وطبيعته وتعبيره الصوتي. يبرز هذا المشروع إمكانية استخدام نماذج *TTS* المتقدمة متعددة اللغات مثل *XTTS* لخدمة اللغات واللهجات محدودة الموارد، مما يمهد الطريق لتقنيات كلام أكثر شمولاً.

الكلمات المفتاحية: اللهجة الجزائرية، *XTTS*، أحدث التقنيات، تركيب الكلام، التعلم بالنقل، محدود الموارد

Contents

Dedication	i
Acknowledgements	ii
Abstract	iii
Résumé	iv
v	ملخص
List of Tables	ix
List of Figures	xi
List of Abbreviations	xiii
Introduction	2
1 Background	2
2 Problem Statement	2
3 Objectives	3
1 Background & Related Works	5
1.1 Chapter Introduction	5
1.2 Fundamental Concepts in Speech Synthesis	6
1.2.1 Artificial Intelligence, Machine Learning, Deep Learning, and Generative Artificial Intelligence	6
Artificial Intelligence (AI)	6
Machine Learning (ML)	6
Deep Learning (DL)	7
Generative Artificial Intelligence (GenAI)	7
1.2.2 The Evolution of Text-to-Speech (TTS)	8
1.3 Key Deep Learning Architectures for Modern TTS	9
1.3.1 Convolutional Neural Networks (CNNs)	9

1.3.2	Recurrent Neural Networks (RNNs)	9
1.3.3	Transformers	10
1.3.4	Generative Adversarial Networks (GANs)	11
1.3.5	Diffusion Models	11
1.4	The Coqui XTTS Model	12
1.4.1	Overview and Key Features	12
1.4.2	Architectural Components	12
1.5	Fine-tuning for Dialect Adaptation	13
1.5.1	The Principle of Fine-tuning	13
1.5.2	Application to Low-Resource Speech Synthesis	13
1.6	Comparative Analysis of Arabic Text-to-Speech Systems	14
1.7	Proposed Approach Based on Related Work	18
1.8	Chapter Summary	19
2	Methodology	20
2.1	Introduction	20
2.2	Contribution: Methodological Approach	20
2.3	Detailed Methodology	21
2.3.1	Data Acquisition	22
2.3.2	Data Preprocessing Pipeline	23
	Initial Audio Chunking Script (Base Model)	24
	Enhanced Audio Chunking Strategy (Final Model)	25
	Audio Duration and Character Limit Considerations	25
2.3.3	Model Fine-tuning Configuration	31
	Base Model Fine-tuning Setup	32
	Final Model Fine-tuning Setup (Projected)	33
	Final Model Fine-tuning Setup (Projected)	33
2.3.4	Implementation and Demonstration Interface	34
2.4	Chapter Conclusion	36
3	Results and Discussion	37
3.1	Introduction	37
3.2	Base Model Fine-tuning Results (12-hour dataset)	37
3.2.1	Quantitative Evaluation: Loss Metrics and Progression	37
3.2.2	Qualitative Evaluation	40
3.2.3	Preprocessing Impact on Base Model Training Data	41
3.3	Results for the Final Model (Projected)	41
3.4	Discussion of Challenges and Limitations	42
3.5	Future Work Considerations	43
3.6	Chapter Conclusion	43
	Conclusion	45

Bibliography

List of Tables

1.1	Comparison of Arabic TTS Systems	18
2.1	Summary of YouTube speakers and audio data duration collected.	22
2.2	Comparison of Data Acquisition for Base and Final Models.	23
3.1	Summary of Quantitative Evaluation Metrics for Base Model Fine-tuning.	38

List of Figures

1.1	Machine Learning Pipeline.	6
1.2	Deep Network Architecture with Multiple Layers.	7
1.3	Generative Artificial Intelligence (GenAI) Concept.	8
1.4	Convolutional Neural Networks (CNNs) Example.	9
1.5	Recurrent Neural Networks (RNNs) Unrolled.	10
1.6	Transformer Architecture Overview.	10
1.7	Example of Diffusion Process in F5-TTS.	11
1.8	XTTS Architectural Diagram.	13
1.9	F5 Architecture Diagram – Model Stages and Data Flow (Arabic Text Support)	16
2.1	Proposed workflow of the project, detailing the methodological stages from data acquisition to model fine-tuning.	21
2.2	dBFS over Time for a 100-second Audio Segment Showing Volume Fluctuations Relevant for Silence Detection and Audio Chunking.	24
2.3	Audio Duration Distribution in the Base Model Dataset.	25
2.4	Audio Duration Distribution in the Final Model Dataset.	26
2.5	Interface Shows Transcript-Audio Synchronization in Real Time.	26
2.6	The word-to-audio length ratio for the base model.	28
2.7	The word-to-audio length ratio for the final model.	28
2.8	Arabic Letters and Corresponding IPA Phonetic Symbols for Phonemization.	30
2.9	Example-Based Correction Rules for Arabic Transcripts used in the AI-assisted refinement stage.	31
2.10	Workflow of the Automated Data Preprocessing Pipeline for Algerian Arabic TTS.	31
2.11	Gradient accumulation computation principle used during model training.	33
2.12	Interactive Algerian Arabic TTS.	34
2.13	Entering Text for Real-Time Synthesis.	35

2.14	Live Audio with Dynamic Text Visualization.	35
3.1	Overall loss progression during fine-tuning, showing Average Mel CE Loss, Average Total Loss, and Average Text CE Loss over 12 cumulative epochs. The significant decrease in Mel CE Loss indicates substantial improvement in audio feature generation.	39
3.2	Detailed progression of Average Text CE Loss. While showing a slight decrease, its values remained relatively low and stable, suggesting the base model's text processing capabilities were already strong or quickly adapted.	39

List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
GenAI	Generative Artificial Intelligence
TTS	Text-to-Speech
XTTS	Cross-lingual Text-to-Speech (Coqui XTTS)
RNN	Recurrent Neural Network
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
MOS	Mean Opinion Score
SRT	SubRip Subtitle

Introduction

1 Background

Text-to-Speech (TTS) technology has revolutionized the way we interact with digital content by converting written text into spoken words. Its applications span various domains, including accessibility tools for individuals with visual impairments, virtual assistants, e-learning platforms, and customer service bots. The global TTS market has witnessed significant growth in recent years. In 2023, it was valued at approximately USD 4 billion and is projected to reach USD 7.6 billion by 2029, growing at a compound annual growth rate (CAGR) of 13.7% [1]. This surge is driven by advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), which have enhanced the naturalness and intelligibility of synthesized speech. In the educational sector, TTS tools have been instrumental in supporting students with reading difficulties, improving word recognition, and enhancing comprehension [2]. Moreover, businesses leverage TTS to provide multilingual support, ensuring inclusivity and expanding their reach to diverse audiences.

2 Problem Statement

Despite the advancements in TTS technology, there remains a significant gap in systems tailored to specific dialects, particularly the Algerian dialect. Most existing TTS systems are designed for widely spoken languages and often overlook regional linguistic nuances. This lack of representation hinders the accessibility and user experience for native Algerian dialect speakers, especially in digital platforms and educational resources.

3 Objectives

The primary objectives of this project are:

- **Development of a TTS Model:** To design and implement a TTS system that accurately synthesizes speech in the Algerian dialect.
- **Dataset Compilation:** To collect and preprocess a comprehensive dataset comprising Algerian dialect text and corresponding audio recordings.
- **Model Evaluation:** To assess the performance of the developed TTS model using objective metrics and subjective listening tests.
- **Accessibility Enhancement:** To contribute to the inclusivity of digital content by providing a tool that caters to Algerian dialect speakers.

4 Report Structure

This report details the development and evaluation of a Text-to-Speech (TTS) model for the Algerian dialect. The **Introduction** (this section) establishes the project's context, problem, objectives, and this report outline.

The subsequent chapters are structured as follows:

- **Chapter 1: Background & Related Works** reviews foundational AI/TTS concepts, key technologies including the Coqui XTTS model, fine-tuning principles, and relevant existing Arabic TTS systems.
- **Chapter 2: Methodology, Implementation, Results, and Evaluation** presents the data acquisition, extensive preprocessing pipeline, fine-tuning configuration, implementation details, and a comprehensive evaluation of the developed TTS model's performance.
- **Chapter 3: Conclusion and Future Work** summarizes the project, assesses objective fulfillment, discusses contributions and limitations, and outlines potential future research directions.

Chapter 1

Background & Related Works

1.1 Chapter Introduction

This chapter lays the groundwork for understanding the development of a Text-to-Speech (TTS) model tailored for the Algerian dialect. We begin by exploring the foundational concepts of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Generative Artificial Intelligence (GenAI), which are pivotal to modern speech synthesis. We then trace the evolution of TTS technology, from early rule-based and statistical methods to the sophisticated neural network-based approaches prevalent today. A significant portion of this chapter is dedicated to dissecting the key DL architectures that underpin state-of-the-art TTS systems, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers, Generative Adversarial Networks (GANs), and Diffusion Models. Building on this, we introduce the Coqui XTTS model, a powerful multilingual and voice-cloning TTS system, detailing its architecture and capabilities. The concept of fine-tuning is then discussed as a crucial technique for adapting large pre-trained models, like XTTS, to specific languages or dialects with limited resources, such as Algerian Arabic. Finally, we review existing research and related works in the field of TTS for Arabic dialects and the fine-tuning of large speech models, identifying current advancements and research gaps. This comprehensive overview will provide the necessary context for the methodology and implementation detailed in subsequent chapters, aimed at creating a high-quality TTS system for the Algerian dialect.

1.2 Fundamental Concepts in Speech Synthesis

1.2.1 Artificial Intelligence, Machine Learning, Deep Learning, and Generative Artificial Intelligence

Artificial Intelligence (AI)

AI is the broadest field, encompassing the theory and development of computer systems able to perform tasks that typically require human intelligence.

Machine Learning (ML)

ML is a subfield of AI that provides systems the ability to automatically learn and improve from experience (i.e., data) without being explicitly programmed for each task [3]. ML algorithms build a model based on sample data, known as "training data," in order to make predictions or decisions without being explicitly programmed to do so. The typical ML pipeline is illustrated in Figure 1.1.

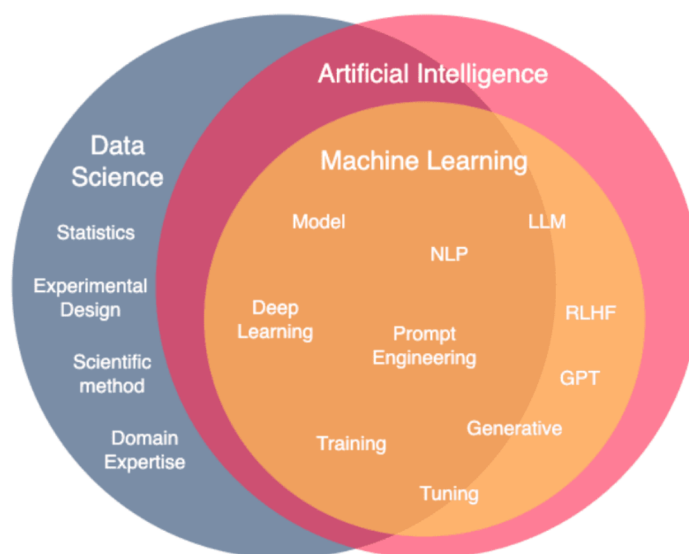


Figure 1.1: Machine Learning Pipeline.

Deep Learning (DL)

DL is a further specialization within ML that utilizes Artificial Neural Networks (ANNs) with multiple layers (hence "deep") to model and understand complex patterns in large datasets [4], as depicted in Figure 1.2. These layers progressively extract higher-level features from the input data. DL has driven significant breakthroughs in various fields, including computer vision, Natural Language Processing (NLP), and, crucially for this project, speech synthesis.

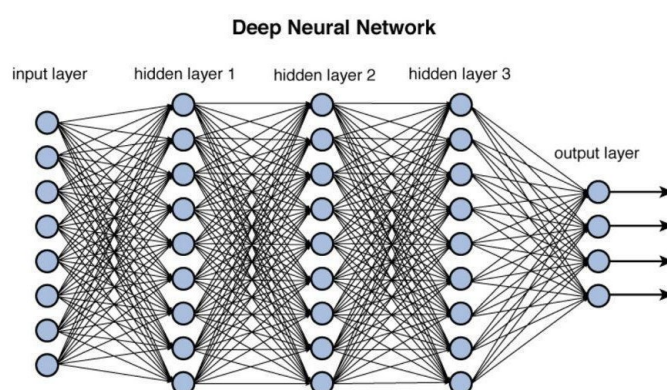


Figure 1.2: Deep Network Architecture with Multiple Layers.

Generative Artificial Intelligence (GenAI)

A significant outcome of advancements in DL is the rise of GenAI. GenAI systems are designed to create new, original content—such as text, images, music, or, as in the focus of this work, speech—rather than simply analyzing or classifying existing data [5]. A conceptual overview of GenAI is shown in Figure 1.3. Modern TTS is a prominent application of GenAI, where models learn the underlying patterns and characteristics of human speech to synthesize novel audio waveforms from textual input. Architectures like GANs and Diffusion Models, discussed later, are core GenAI techniques heavily utilized in state-of-the-art TTS.

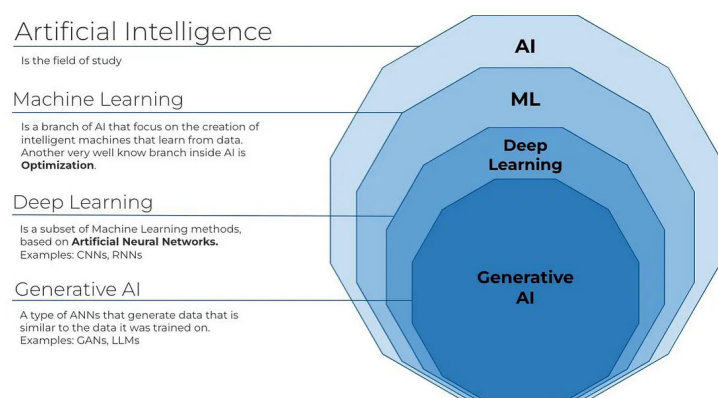


Figure 1.3: Generative Artificial Intelligence (GenAI) Concept.

1.2.2 The Evolution of Text-to-Speech (TTS)

TTS technology aims to convert written text into audible human-like speech. Over the decades, TTS systems have evolved significantly, driven by advancements in computational power and algorithmic sophistication.

Concatenative and Parametric TTS Early TTS systems primarily used two approaches: concatenative synthesis and parametric synthesis. Concatenative TTS involved stitching together pre-recorded speech units (e.g., phonemes, diphones, or even words) from a large database. While capable of producing natural-sounding speech for common phrases, it often suffered from audible discontinuities at the joins of concatenated units and lacked flexibility in modifying voice characteristics [6]. Parametric TTS, often employing Statistical Parametric Speech Synthesis (SPSS) with Hidden Markov Models (HMMs), modeled the statistical relationship between linguistic features and acoustic parameters (e.g., fundamental frequency, spectral envelope). These parameters were then used to drive a vocoder to generate the speech waveform. While more flexible and requiring smaller footprints, HMMs-based TTS often produced speech that sounded “buzzy” or “muffled” [7].

Neural TTS The advent of DL marked a paradigm shift, leading to Neural TTS. These systems employ deep neural networks to directly map input text (or phoneme sequences) to acoustic representations (typically mel-spectrograms). Another neural network, known as a neural vocoder, then converts these mel-spectrograms into high-fidelity audio waveforms. Neural TTS models, such as Google’s Tacotron [8] and Baidu’s Deep Voice [9],

have demonstrated a remarkable ability to produce speech that is significantly more natural, expressive, and human-like than previous methods.

1.3 Key Deep Learning Architectures for Modern TTS

1.3.1 Convolutional Neural Networks (CNNs)

CNNs, while originally famed for image processing, have also proven effective for sequence modeling tasks, including TTS [13]. An example can be seen in Figure 1.4. 1D CNNs can capture local contextual information from text sequences or temporal patterns in speech features. They are often used for feature extraction or as components within larger architectures like Tacotron 2 [14], which combines CNNs and LSTMs.

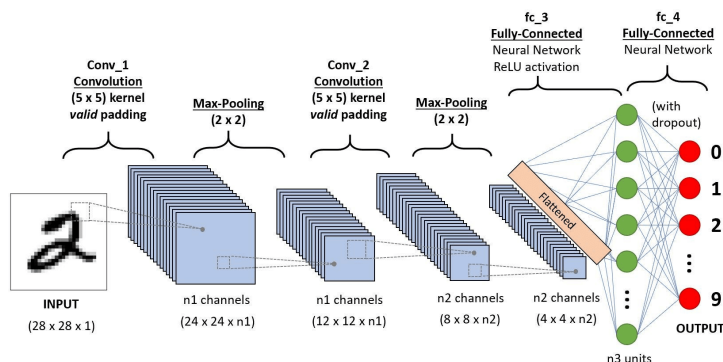


Figure 1.4: Convolutional Neural Networks (CNNs) Example.

1.3.2 Recurrent Neural Networks (RNNs)

RNNs are designed to process sequential data, making them naturally suited for text and speech [10]. An unrolled RNN is visualized in Figure 1.5. Variants like Long Short-Term Memory (LSTM) [11] and Gated Recurrent Units (GRU) [12] incorporate gating mechanisms to effectively learn long-range dependencies, mitigating the vanishing/exploding gradient problems of simple RNNs. LSTMs and GRUs have been widely used in the encoder and decoder components of early attention-based neural TTS systems like Tacotron.

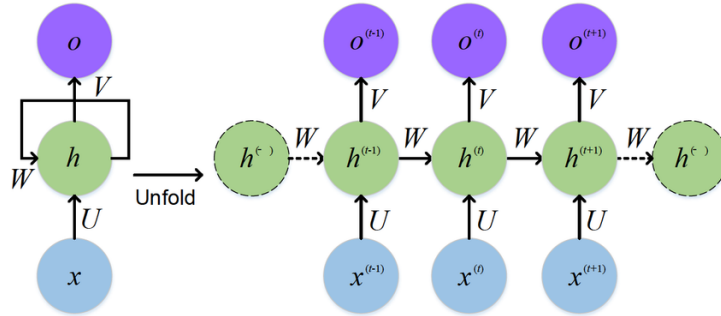


Figure 1.5: Recurrent Neural Networks (RNNs) Unrolled.

1.3.3 Transformers

The Transformer architecture, introduced in “Attention Is All You Need” [15], revolutionized sequence modeling. Its overview is presented in Figure 1.6. It relies entirely on self-attention mechanisms to draw global dependencies between inputs and outputs, dispensing with recurrence and allowing for significant parallelization during training. Transformers excel at capturing long-range context in text (for the text encoder part of TTS) and can also be used in the decoder to generate acoustic features. Many state-of-the-art TTS models, such as Transformer TTS [16] and FastSpeech [17], heavily leverage the Transformer architecture.

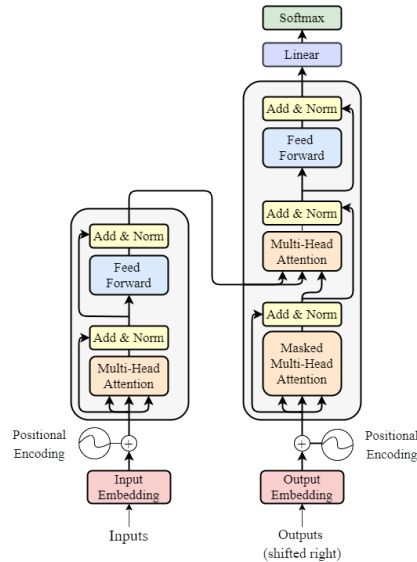


Figure 1.6: Transformer Architecture Overview.

1.3.4 Generative Adversarial Networks (GANs)

GANs [18] involve two neural networks, a Generator and a Discriminator, trained in a competitive setting. The Generator learns to produce data (e.g., audio waveforms) that resembles real data, while the Discriminator learns to distinguish between real and generated data. In TTS, GANs are predominantly used in neural vocoders to convert mel-spectrograms into highly realistic audio waveforms. Examples include MelGAN [19] and HiFi-GAN [20], the latter being known for its high fidelity and efficiency.

1.3.5 Diffusion Models

Diffusion Probabilistic Models (or simply Diffusion Models) are a class of generative models that have recently achieved state-of-the-art results in image and audio generation [21]. They work by learning to reverse a gradual noising process. Starting from pure noise, the model iteratively refines the data, conditioned on some input (like text embeddings and speaker identity), to generate a clean sample (e.g., a mel-spectrogram). The diffusion process is exemplified in Figure 1.7. Flow Matching [22] is a related, newer technique for training continuous normalizing flows that can offer training efficiency. Diffusion models are used in TTS acoustic models like Grad-TTS [23] and are a key component in recent versions of Coqui XTTS for generating high-quality and expressive speech.

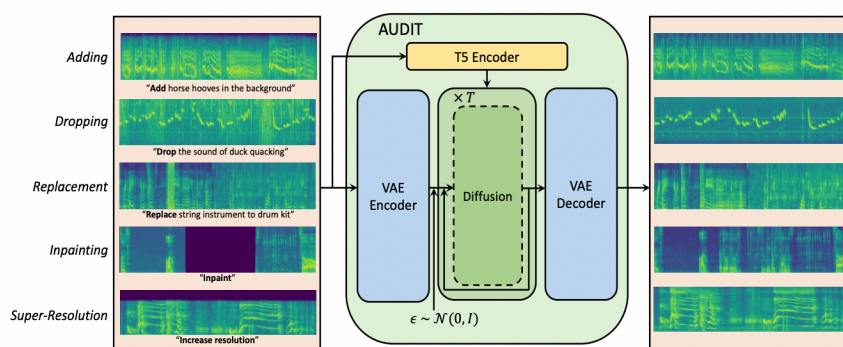


Figure 1.7: Example of Diffusion Process in F5-TTS.

1.4 The Coqui XTTS Model

1.4.1 Overview and Key Features

Coqui XTTS is an open-source, DL-based TTS model that stands out for its multilingual capabilities and impressive zero-shot voice cloning [24]. Zero-shot voice cloning means the model can synthesize speech in a target voice using only a few seconds of reference audio from that speaker, even if the speaker was not seen during the model's initial training. XTTS aims to produce highly natural and expressive speech.

1.4.2 Architectural Components

While specific architectural details can evolve, XTTS models (particularly versions like XTTS v2) generally follow a modern neural TTS pipeline, likely incorporating elements shown in Figure 1.8:

- **Text Encoder:** Processes the input text (often after phonemization) and converts it into a sequence of contextual embeddings. This component is typically Transformer-based to effectively capture linguistic information and long-range dependencies.
- **Speaker Encoder:** Extracts a speaker embedding from a reference audio clip. This embedding captures the unique vocal characteristics (timbre, pitch, etc.) of the target voice, enabling voice cloning.
- **Variance Adaptor (or similar mechanism):** This component, often found in non-autoregressive TTS models, helps predict and model prosodic variations like duration, pitch, and energy, contributing to more natural-sounding speech aligned with the text.
- **Acoustic Model/Decoder:** This is the core generative part that synthesizes acoustic features (e.g., mel-spectrograms) from the text embeddings, speaker embedding, and prosody information. In recent XTTS versions, this is often a Diffusion-based model due to its ability to generate high-quality and diverse outputs.
- **Vocoder:** Converts the generated mel-spectrograms into an audible waveform. Coqui often utilizes a GANs-based vocoder like HiFi-GAN for this stage to ensure high audio fidelity.

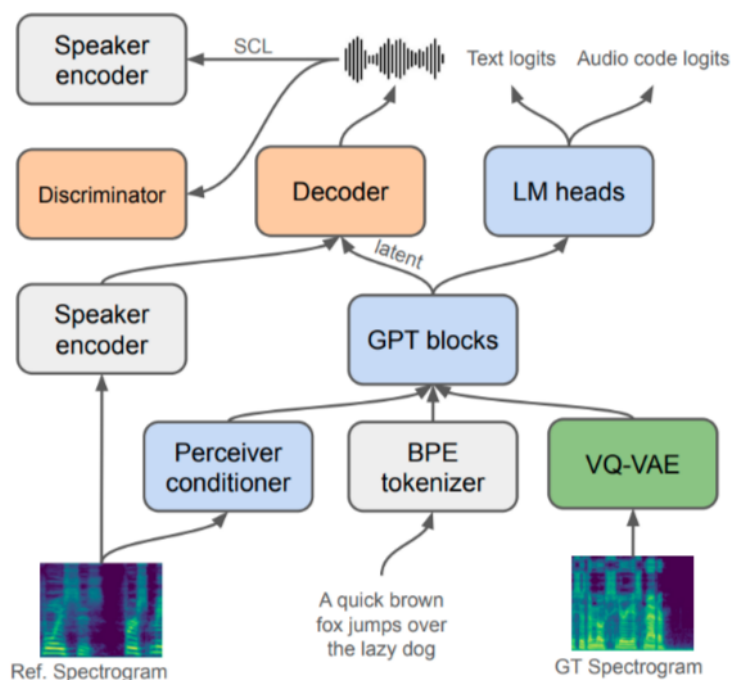


Figure 1.8: XTTS Architectural Diagram.

1.5 Fine-tuning for Dialect Adaptation

1.5.1 The Principle of Fine-tuning

Fine-tuning is a transfer learning technique where a model pre-trained on a large, general dataset (e.g., diverse languages and speakers) is further trained on a smaller, task-specific dataset [25]. Instead of training a complex model from scratch, which is computationally expensive and requires vast amounts of data, fine-tuning adapts the learned knowledge of the pre-trained model to the nuances of the new, specific data. This typically involves unfreezing some or all of the pre-trained model’s layers and continuing the training process with a lower learning rate.

1.5.2 Application to Low-Resource Speech Synthesis

Many languages and dialects, including Algerian Arabic, are considered “low-resource” in terms of publicly available, high-quality speech datasets. Training a state-of-the-art TTS model from scratch for such a dialect would be extremely challenging. Fine-tuning a powerful pre-trained multilingual model like XTTS offers a viable and efficient solution. By fine-tuning on a

relatively small dataset of Algerian dialect speech, the model can adapt its existing knowledge of phonetics, prosody, and language structure to accurately pronounce Algerian words, adopt dialect-specific intonation, and generate natural-sounding speech in the target dialect.

1.6 Comparative Analysis of Arabic Text-to-Speech Systems

This section presents a detailed comparison of recent Arabic TTS systems, focusing on their architectural designs, training strategies, datasets, performance metrics, and their respective strengths and weaknesses.

1. EGTTS-V0.1 – Egyptian Arabic TTS by Omar Samir et al.

- **Architecture:** Built upon the XTTS v2 framework, which supports multilingual voice cloning and synthesis.
- **Training Dataset:** Fine-tuned using 17 hours of Egyptian Arabic audio data.
- **Training Details:** Leveraged the pre-trained XTTS v2 model, adapting it to the Egyptian Arabic dialect through fine-tuning.
- **Strengths:**
 - Tailored specifically for Egyptian Arabic, capturing dialectal nuances.
 - Utilizes the robust XTTS v2 architecture for high-quality synthesis.
- **Weaknesses:**
 - Limited training data may affect generalization to broader contexts.
- **Performance:** Demonstrated natural-sounding speech synthesis for Egyptian Arabic. [30]

2. Orpheus-TTS – Multilingual TTS with Paralinguistic Features

- **Architecture:** Based on the Llama-3b backbone, designed for high-quality, empathetic text-to-speech generation.
- **Training Dataset:** Pre-trained on over 100,000 hours of multilingual data, primarily in English.
- **Training Details:** While the core model is extensively trained, fine-tuning for Arabic is limited.
- **Strengths:**
 - Supports expressive elements like <laugh>, <sigh>, enhancing naturalness.
 - Capable of real-time streaming performances with low latency.
- **Weaknesses:**
 - Limited fine-tuning for Arabic may affect performance in that language.
- **Performance:** Achieves exceptional clarity and expressiveness in supported languages; Arabic performance may vary. [31]

3. F5-TTS – High-Quality Arabic Speech Synthesis

- **Architecture:** Utilizes a non-autoregressive Diffusion Transformer (DiT) architecture with a ConvNeXt-based text encoder.
- **Training Dataset:** Fine-tuned on the MBZUAI/CIArTTS dataset, focusing on Modern Standard Arabic (MSA).
- **Training Details:** Ongoing fine-tuning with temporary checkpoints; the latest at 380,000 steps.
- **Strengths:**
 - Emphasizes regional pronunciation diversity.
 - Achieves a Real-Time Factor (RTF) of 0.15, allowing efficient deployment.
- **Weaknesses:**

- Limited dataset size may impact the model's ability to generalize across different dialects.
- **Performance:** Demonstrates strong performance with clear pronunciation and natural prosody in MSA. [32]

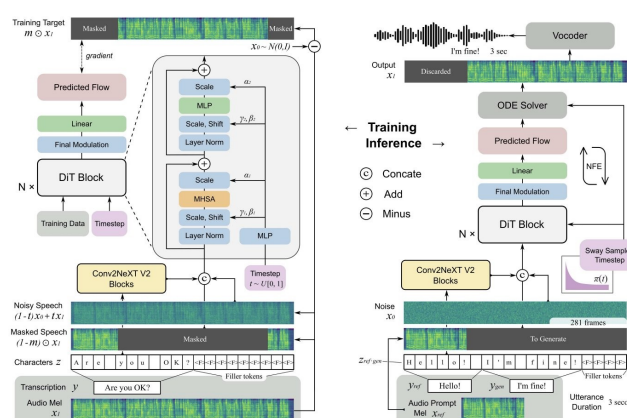


Figure 1.9: F5 Architecture Diagram – Model Stages and Data Flow (Arabic Text Support)

4. X-TTS – Cross-Lingual Voice Cloning

- **Architecture:** A multilingual voice cloning system capable of generating speech from a 3-second reference audio.
- **Training Dataset:** Trained on a diverse multilingual dataset supporting 16 languages, including Arabic.
- **Training Details:** Designed for rapid prototyping and low-data adaptation without extensive language-specific training.
- **Strengths:**
 - Enables cross-lingual synthesis, beneficial for under-resourced dialects.
 - Requires minimal reference audio for voice cloning.
- **Weaknesses:**

- Performance may vary depending on the quality and length of the reference audio.
- **Performance:** Capable of generating lifelike voice synthesis across supported languages. [33]

5. Zero-Shot TTS for Arabic Dialects

- **Architecture:** Fine-tuned the XTTS model on a curated Arabic corpus with embedded dialect labels.
- **Training Dataset:** Utilized the QASR dataset, containing approximately 2,000 hours of speech from Aljazeera news, covering multiple dialects.
- **Training Details:** Employed Arabic dialect identification models to improve the zero-shot TTS model in a multi-dialect setting.
- **Strengths:**
 - Capable of synthesizing dialectal Arabic without extensive per-dialect retraining.
 - Demonstrated strong intelligibility and dialectal accuracy.
- **Weaknesses:**
 - Performance may vary across less-represented dialects in the training data.
- **Performance:** Achieved convincing performance in generating dialectal speech, evaluated across 31 unseen speakers. [34]

The following table presents a comparison of various TTS models based on key performance metrics.

Model	Architecture	Dataset	Key Strengths
EGTTS-V0.1	XTTS v2	17 hours Egyptian Arabic	Dialect-specific synthesis
Orpheus-TTS	Llama-3b	100k+ hours (primarily English)	Expressiveness, real-time performance
F5-TTS	DiT with Con-vNeXt	MBZUAI/CIArTTS (MSA)	Regional diversity, efficient deployment
X-TTS	Multilingual voice cloning	16 languages including Arabic	Cross-lingual synthesis, minimal data requirement
Zero-Shot TTS	XTTS fine-tuned with dialect labels	QASR (2,000 hours)	Multi-dialect synthesis without re-training

Table 1.1: Comparison of Arabic TTS Systems

1.7 Proposed Approach Based on Related Work

Building upon the insights from previous studies, this work will utilize the Coqui XTTS model, similar to the approach taken by the EGTTS-V0.1 project [30] and the Zero-Shot TTS for Arabic Dialects [34]. The goal is to achieve high-quality voice synthesis for the Algerian dialect while using a relatively small, carefully curated dataset, addressing the common challenge of limited data availability for specific dialects.

Additionally, this project will leverage the inherent voice cloning capabilities of XTTS [24] by fine-tuning on specific speaker data within our Algerian Arabic dataset, aiming to replicate speaker characteristics accurately. To enhance the naturalness and emotional impact of the synthesized speech, expressiveness modeling, potentially guided by prosodic features or emotion tags identified during preprocessing (as will be detailed in Chapter 2), will also be a focus. By combining these strategies—fine-tuning a powerful pre-trained model, meticulous dataset curation, and attention to voice cloning and expressiveness—the system aims to produce clear, natural, and engaging speech output for Algerian Arabic with efficient data usage, contributing a valuable resource for this under-represented dialect.

1.8 Chapter Summary

This chapter has provided a comprehensive overview of the technological landscape relevant to developing a TTS system for the Algerian dialect. We started with the fundamental concepts of AI, ML, DL, and GenAI, traced the evolution of TTS technologies, and detailed the key deep learning architectures—RNNs, CNNs, Transformers, GANs, and Diffusion Models—that enable modern, high-quality speech synthesis. The Coqui XTTS model was introduced as a powerful platform for this project, and the principle of fine-tuning was explained as the core methodology for adapting XTTS to the target dialect. The review of related works highlighted existing efforts in Arabic TTS and fine-tuning. The subsequent discussion on our proposed approach, informed by these related works, outlined the specific strategies for this project, underscoring the need for research on Algerian Arabic. This foundational knowledge paves the way for the subsequent chapters, which will detail the dataset creation, fine-tuning process, and evaluation of the developed Algerian dialect TTS model.

Chapter 2

Methodology

2.1 Introduction

This chapter details the comprehensive methodology employed to develop and fine-tune a Text-to-Speech (TTS) model for the Algerian Arabic dialect using the Coqui XTTS-v2 framework. The primary objective is to generate high-quality, natural-sounding speech by meticulously preparing dialect-specific data and systematically configuring the model. We will describe the data acquisition strategy, the extensive automated preprocessing pipeline, and the specific configurations for fine-tuning both an initial base model and a projected final model.

2.2 Contribution: Methodological Approach

The core contribution of this work, detailed in the subsequent sections of this chapter, lies in the specific methodology employed to develop a Text-to-Speech (TTS) model for the Algerian dialect. This methodology systematically addresses the challenges of working with a low-resource language variant by implementing a structured, multi-stage process.

Our approach begins with a targeted **Data Acquisition** strategy, focusing on sourcing relevant Algerian Arabic speech content and initial transcripts from online platforms, with specific criteria for speaker characteristics and data recency. Following acquisition, the raw data undergoes an extensive and largely automated **Data Preprocessing Pipeline**. This crucial phase encompasses several intricate steps: audio segmentation, precise transcript alignment, content and quality-based filtering, audio enhancement, phonemization, and AI-assisted transcript correction.

Once a high-quality, curated dataset is prepared, we proceed to the **Model Fine-tuning** stage. This involves adapting the pre-trained Coqui XTTS-v2 model through a two-stage process: first, training a base model on an initial dataset, and subsequently, training a final, more robust model by further refining the best-performing checkpoint from the base model using a significantly larger dataset. The specific configurations and parameters for both fine-tuning stages are presented herein. This methodological framework forms the foundation for creating the Algerian dialect TTS system.

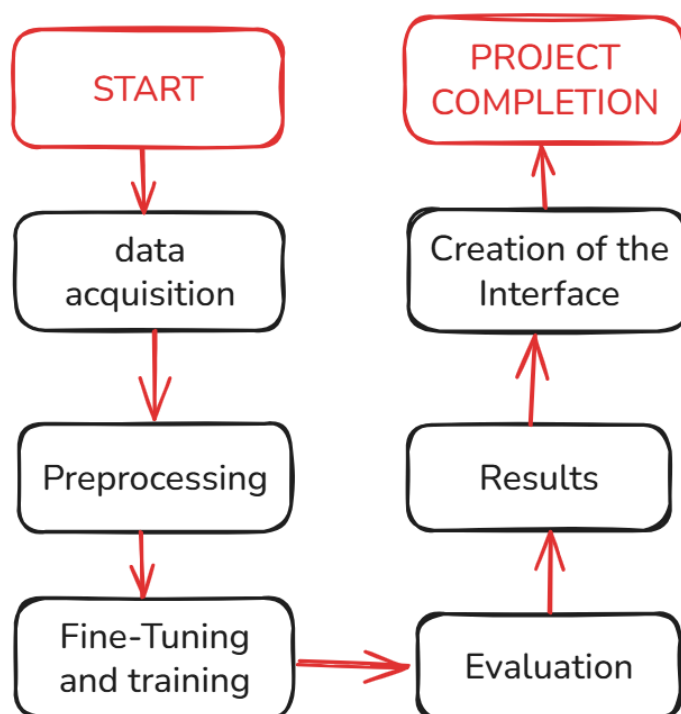


Figure 2.1: Proposed workflow of the project, detailing the methodological stages from data acquisition to model fine-tuning.

2.3 Detailed Methodology

The overall methodology follows a two-stage fine-tuning approach. First, a base model was fine-tuned using an initial dataset of approximately 12 hours of processed Algerian Arabic speech. Subsequently, a final model is

planned, which will involve further fine-tuning the best checkpoint from the base model stage with an expanded dataset exceeding 100 hours, following more advanced optimization and enhancement procedures. This progressive strategy aims to leverage the benefits of transfer learning while adapting the model to better represent the nuances of a low-resource dialect through increasingly rich and refined data.

2.3.1 Data Acquisition

The data acquisition strategy was consistent for both the base and final model datasets.

- **Source:** YouTube videos featuring Algerian Arabic speech were the primary data source.

Speaker / Folder	Total Duration (hours)	Reference
AymanCha7el	37.5	[35]
Aymancha7el2	0.49	[36]
AyoubPeter	9.58	[37]
basair	2.31	[38]
chawech	2.79	[39]
loubna_1	15.19	[40]
loubna_2	50.35	[41]
OmarRahmoun09	7.12	[42]
salmi_mohamed	4.04	[43]
Total	129.37	

Table 2.1: Summary of YouTube speakers and audio data duration collected.

- **Recency Filter:** For the base model, only videos uploaded within the two months prior to collection were selected. This was done to leverage recent improvements in YouTube’s automatic transcription, particularly for dialectal content with loanwords. No recency constraint was applied for the final model.
- **Initial Collection Volume:**
 - **Base Model:** Approximately 22 hours of raw audio-visual material were collected. After preprocessing, 12 hours of usable single-speaker data were retained.

- **Final Model:** Approximately 130 hours of raw data were collected. After preprocessing, 110 hours of clean, single-speaker audio segments were retained.
- **Speaker Segmentation:**
 - **Base Model:** Some audio segments included multiple speakers in a single file.
 - **Final Model:** Each audio file contains speech from only one speaker to ensure cleaner training data.
- **Transcription Source:** YouTube’s auto-generated SRT (SubRip Subtitle) files were extracted and used as the initial transcripts for all audio segments.

The following table provides a detailed comparison between the data acquisition strategies used for the base and final models:

Aspect	Base Model	Final Model
Source	YouTube videos with Algerian Arabic speech	YouTube videos with Algerian Arabic speech
Recency Filter	Only videos uploaded within 2 months prior to collection (to leverage recent YouTube transcription improvements)	No recency constraint applied
Initial Collection Volume	Approximately 22 hours of raw audio-visual material collected	Approximately 130 hours of raw data collected
Post-Preprocessing Usable Data	12 hours of clean, single-speaker audio retained	110 hours of clean, single-speaker audio retained
Speaker Segmentation	Some audio files contained multiple speakers; speakers randomly selected between male and female	Each audio file contains speech from only one speaker; balanced 65 hours for male and female speakers
Transcription Source	YouTube auto-generated SRT subtitles used as initial transcripts	YouTube auto-generated SRT subtitles used as initial transcripts

Table 2.2: Comparison of Data Acquisition for Base and Final Models.

2.3.2 Data Preprocessing Pipeline

A comprehensive, automated preprocessing pipeline was implemented to prepare the collected data for fine-tuning. This pipeline was used for the base model dataset and will be replicated for the additional data for the final model. The key stages are outlined below:

1. Audio Chunking and Initial Filtering:

Chunking—splitting recordings into smaller segments—is a key step in preparing audio for TTS training. It ensures words aren’t cut off mid-speech, keeps segment lengths consistent, and improves alignment with transcripts. Two script versions were used: one for the base dataset and an enhanced version for the final, larger dataset.

Initial Audio Chunking Script (Base Model)

Audio segmentation for the base model dataset was done with a custom Python script using Pydub, refining chunks to 3–30 seconds. Processing 1 hour of audio took about 12 minutes. Key steps included:

1. Initial detection of non-silent regions using Pydub’s silence detection with an adaptive threshold.
2. A short amount of padding (200 ms) was applied to detected regions to preserve word boundaries.
3. An iterative refinement loop merged segments that were too short and re-split those that were too long, using progressively adjusted silence detection parameters.
4. The final audio chunks were filtered to ensure a minimum duration of 500 ms, normalized to -20.0 dBFS, padded with 50 ms of silence, and exported as WAV files. A summary Excel report cataloged all generated chunks.

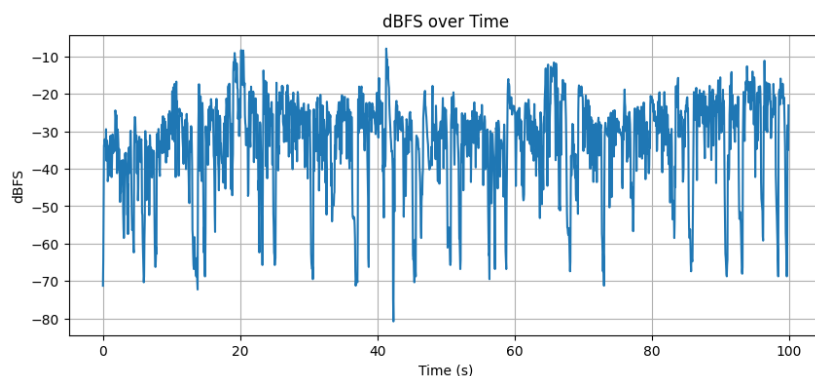


Figure 2.2: dBFS over Time for a 100-second Audio Segment Showing Volume Fluctuations Relevant for Silence Detection and Audio Chunking.

Enhanced Audio Chunking Strategy (Final Model)

To handle audio segments exceeding the maxduration threshold of 20 seconds, a multi-stage chunking strategy was applied:

1. **Silence Detection Near Time Marks:** Within the 20-second window, silence points were searched around predefined time marks (from 1 to 18 seconds) using standard silence parameters (`silencethresh` = -40 dB, `silencelen` = 200 ms), to ensure well-distributed chunk boundaries.
2. **Secondary Silence Detection:** If no silence was found, a second pass with more sensitive parameters (`silencethresh` = -35 dB, `silencelen` = 130 ms) was performed across the full 20-second segment.
3. **Forced Split:** If silence was still not detected, the segment was forcibly split at exactly maxduration (20 seconds).

The predefined time marks are used cyclically to improve chunk distribution, with volume normalization and brief silence padding applied to each resulting chunk to ensure consistency and audio quality.

Audio Duration and Character Limit Considerations

The **base model dataset** consisted mostly of short audio segments ranging from 3 to 30 seconds. Due to the model's constraint of not accepting Arabic inputs longer than 166 characters, approximately 30% of samples were discarded before training. This filtering helped avoid training errors but led to data loss.

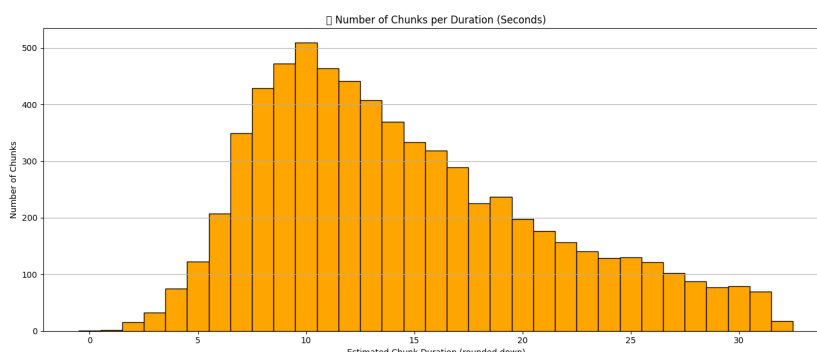


Figure 2.3: Audio Duration Distribution in the Base Model Dataset.

For the **final model**, improved chunking logic produced a more balanced duration distribution between 2 and 20 seconds, minimizing overly short or long chunks. This alignment ensured most chunks met the text length requirement, reducing data loss and enhancing overall training quality.

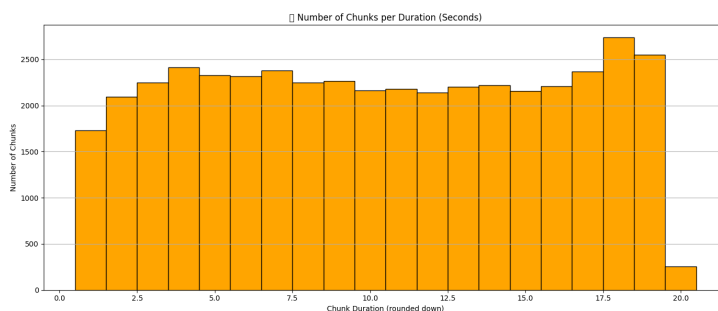


Figure 2.4: Audio Duration Distribution in the Final Model Dataset.

2. Transcript Association and Refinement:

- **SRT Parsing and Alignment:** Corresponding SRT files were used to extract transcripts.
- **Precise Transcript Alignment:** A custom function, employing word-level timing estimation within subtitles, was utilized to accurately map transcript segments to the precise start and end times of each audio chunk. This was crucial as subtitle timings often do not perfectly align with automatically generated audio chunk boundaries.



Figure 2.5: Interface Shows Transcript-Audio Synchronization in Real Time.

3. Word Rate Filtering

To ensure high-quality training data, we applied a filtering step based on the speech rate of each audio chunk, measured in Words Per Second (WPS). This step aimed to remove segments with unnaturally slow speech or silence, which may not be useful for training robust models.

Word Counting: Each transcript was processed using an Arabic-specific word counting function to accurately calculate the total number of spoken words.

WPS Calculation: For each chunk, the word rate was computed using the formula:

$$WPS = \frac{WordCount}{Duration(s)}$$

Low WPS Chunk Removal: Chunks with abnormally low WPS values were flagged for removal. A dynamic threshold was used, defined as:

$$LowerThreshold = \max(0.2, \mu_{WPS} - 2\sigma_{WPS})$$

where μ_{WPS} is the average WPS and σ_{WPS} is the standard deviation of WPS. This ensures that only significantly slow or silent segments are excluded, while a minimum threshold of 0.2 WPS prevents over-filtering.

Overall, for the final model dataset, this process led to the removal of approximately 0.5% of the data, equivalent to 1800 chunks. This filtering helped maintain a consistent and meaningful speech rate across the training corpus.

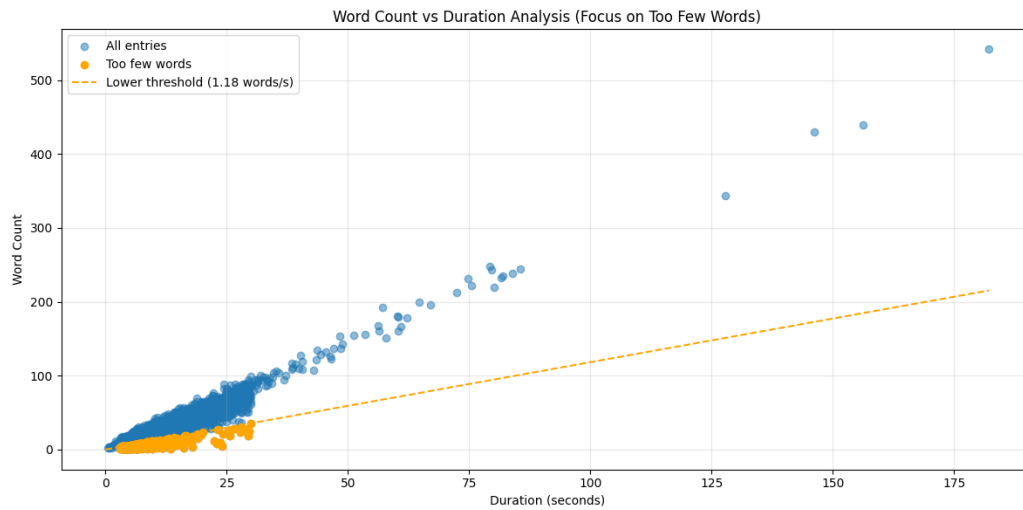


Figure 2.6: The word-to-audio length ratio for the base model.

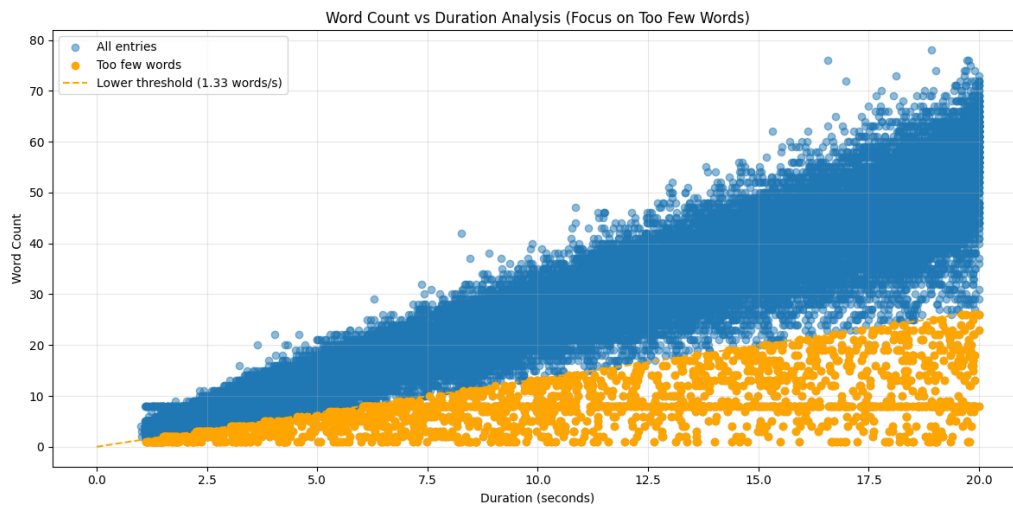


Figure 2.7: The word-to-audio length ratio for the final model.

4. Content-Based Filtering - Explicit Music Tag Removal:

- Transcripts were programmatically scanned for the explicit tag "[موسيقى]" (music). This tag is an annotation often provided by YouTube within its auto-generated SRT files.
- The remaining data, presumed to be primarily speech, formed the input for the next stage.

5. Advanced Audio Analysis and Processing:

- **Tooling:** This stage used PANNs (Large-Scale Pretrained Audio Neural Networks) for audio tagging (music detection) and sound event detection (emotional cues) [44], and DeepFilterNet for denoising [45].
- **Music Detection (PANNs-AT):** Audio chunks were analyzed for music presence with a confidence threshold (e.g., 0.3).
- **Denoising (DeepFilterNet):** If significant music was detected, the audio chunk was processed by DeepFilterNet3 to suppress non-speech sounds. The original file was replaced by its denoised version, with the original archived.
- **Emotional Sound Detection (PANNs-SED):** Tried only in the base model. Emotional sounds (Screaming, Whispering, Laughter) were identified with a confidence threshold (e.g., 0.1), but this method was later dropped due to poor alignment with transcripts.
- **Metadata Update:** The metadata report was updated with music detection flags and denoising status. Emotion-related fields were only included in the base model's trial and not in later versions.

6. Final Duration-Based Filtering:

- **Base Model:** The dataset was refined by removing audio chunks with durations less than 2 seconds or greater than 30 seconds. This filtering step helped standardize segment lengths, which is beneficial for training text-to-speech (TTS) models. The metadata report was updated accordingly.
- **Final Model Dataset:** In the final version of the dataset, advanced chunking techniques were implemented, producing consistently well-sized audio segments. As a result, the duration-based filtering step was deemed unnecessary and was no longer applied.

7. Phonemization:

- Transcripts of the filtered and processed audio chunks were transformed into phonemic representations using a Wav2Vec2-based model (vitouphy/wav2vec2-xls-r-300m-timit-phoneme) [46]. Audio was resampled to 16kHz to match the model's requirements. The mapping between Arabic graphemes and the target IPA symbols is detailed in Table ??.

The following figure provides a visual representation of the mapping between Arabic letters and their corresponding International Phonetic Alphabet (IPA) symbols utilized during the phonemization process.

Arabic	Name	Symbol	Arabic	Name	Symbol
ء	<i>hamzah</i> '[ʔ]'	ʔ	ص	<i>ṣād</i> '[s̥]'	s̥/s̥'/s̥'*
ب	<i>bāʾ</i> '[b]'	b	ض	<i>ḍād</i> '[d̥]'	d̥/d̥'/d̥'***
ت	<i>tāʾ</i> '[t]'	t	ط	<i>ṭāʾ</i> '[t̤]'	t̤/t̤'/t̤'*
ث	<i>ṭhāʾ</i> '[θ]'	θ	ظ	<i>ḍāʾ</i> '[d̥]'	d̥/ð̥'/ð̥'***
ج	<i>gīm</i> '[g]'	g	ع	<i>ʕayn</i> '[ʕ]'	ʕ/ʕ'*
ح	<i>ḥāʾ</i> '[h]'	h/h'*	غ	<i>ḡayn</i> '[ɣ]'	ɣ
خ	<i>xāʾ</i> '[x]'	x	ف	<i>fāʾ</i> '[f]'	f
د	<i>dāl</i> '[d]'	d	ق	<i>qāʾ</i> '[q]'	q/q'*
ذ	<i>ḏāl</i> '[ð]'	ð	ك	<i>kāʾ</i> '[k]'	k
ر	<i>rāʾ</i> '[r]'	r	ل	<i>lām</i> '[l]'	l
ز	<i>zay</i> '[z]'	z	م	<i>mīm</i> '[m]'	m
س	<i>sīm</i> '[s]'	s	ن	<i>nūn</i> '[n]'	n
ش	<i>šīm</i> '[ʃ]'	ʃ/ʃ̣	هـ	<i>hāʾ</i> '[h]'	h

Figure 2.8: Arabic Letters and Corresponding IPA Phonetic Symbols for Phonemization.

8. Transcript Correction using Generative AI (Gemini):

- A final transcript refinement step was designed using Google's Gemini model.
- The model was prompted to analyze batches of sequential transcript segments along with their corresponding phoneme sequences to correct word misplacements (e.g., words at the end of one chunk phonetically belonging to the start of the next, or vice-versa) and remove extraneous words lacking phonemic evidence. The emphasis was on leveraging phonemic evidence for structural corrections while maintaining semantic coherence.

- The output was structured as a JSON array of [original_transcript, corrected_transcript] pairs, which then updated the dataset metadata. This corrected dataset formed the basis for the final `metadata.csv` used in XTTS fine-tuning.

```

javascript
1 const cases = [
2   // Case 1: Word order fix - misplaced "ان"
3   [
4     "صباح الخير كيف الحال ان شاء الله تكون بخير صحي",
5     "صباح الخير كيف الحال ان شاء الله تكون بخير صحي",
6   ],
7
8   // Case 2: Word order fix - 'الحال' was split
9   [
10    "صباح الخير كيف الحال ان شاء الله تكون بخير صحي",
11    "صباح الخير كيف الحال ان شاء الله تكون بخير صحي",
12  ],
13
14  // Case 3: Phrase shift - remove redundant 'كيف الحال'
15  [
16    "صباح الخير كيف الحال ان شاء الله تكون بخير صحي",
17    "صباح الخير ان شاء الله تكون بخير صحي",
18  ],
19 ];

```

Figure 2.9: Example-Based Correction Rules for Arabic Transcripts used in the AI-assisted refinement stage.

The automated data preprocessing pipeline, as described in detail above, follows a series of sequential stages. Figure 2.10 below provides a visual representation of this workflow.

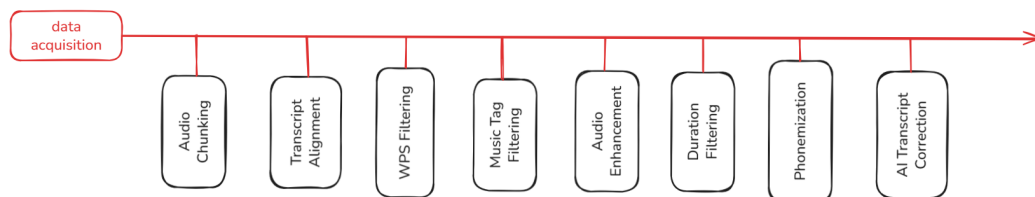


Figure 2.10: Workflow of the Automated Data Preprocessing Pipeline for Algerian Arabic TTS.

2.3.3 Model Fine-tuning Configuration

The fine-tuning process was built upon the Coqui XTTS-v2 model (version 2.0.1), utilizing the Coqui TTS trainer library [47].

Base Model Fine-tuning Setup

- **Hardware:** Fine-tuning was conducted on Google Colab using an NVIDIA A100 GPU with 40 GB VRAM, providing sufficient memory for training large-scale models efficiently.
- **Starting Point:** Fine-tuning began from the official pre-trained Coqui XTTS v2.0.1 checkpoint.
- **Model Arguments (GPTArgs):**
 - max_conditioning_length: 143,677 samples
 - min_conditioning_length: 66,150 samples
 - max_wav_length: 520,000 samples
 - max_text_length: 200 characters
 - pretrained_paths: Included XTTS v2.0.1 components — DVAE, mel normalization, tokenizer, and base model checkpoint
- **Audio Configuration (XttsAudioConfig):**
 - Input/DVAE sample rate: 22,050 Hz
 - Output sample rate: 24,000 Hz
- **Trainer Configuration (GPTTrainerConfig):**
 - Epochs: 15 (with manual interruption expected)
 - Output path: Google Drive
 - Batch size: 4 with 16 gradient accumulation steps (effective batch size: 64)
 - Optimizer: AdamW with a learning rate of 5×10^{-6}
 - Scheduler: MultiStepLR
 - Language: Arabic ("ar")
 - Evaluation: Test sentences were synthesized using a reference speaker WAV for qualitative assessment
- **Dataset Configuration (BaseDatasetConfig):**
 - Format: LJSpeech-style dataset
 - Metadata: A generated metadata.csv file
 - Evaluation split: 2% of the dataset

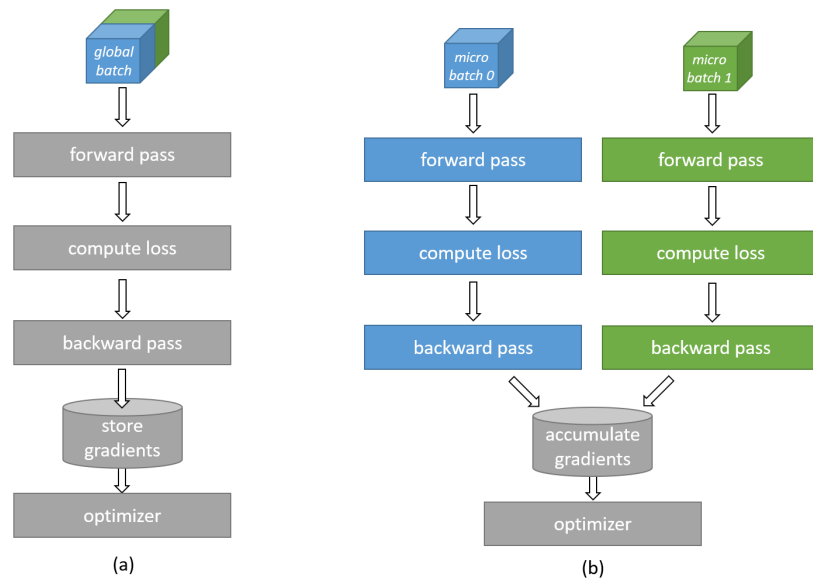


Figure 2.11: Gradient accumulation computation principle used during model training.

Final Model Fine-tuning Setup (Projected)

Fine-tuning for the final model will resume from the best-performing checkpoint from the base model training stage.

- **Checkpoint Initialization:** Weights will be loaded from the base model's best-performing checkpoint.
- **Configuration Adjustments:**
 - Core XTTS architecture and audio settings will remain unchanged.
 - Epochs, batch size, and learning rate schedule may be tuned based on base model results and expanded dataset characteristics.

Final Model Fine-tuning Setup (Projected)

Fine-tuning for the final model will resume from the best-performing checkpoint from the base model training stage.

- **Checkpoint Initialization:** Weights will be loaded from the base model's best-performing checkpoint.
- **Configuration Adjustments:**

- Core XTTS architecture and audio settings will remain unchanged.
- Epochs, batch size, and learning rate schedule may be tuned based on base model results and expanded dataset characteristics.

2.3.4 Implementation and Demonstration Interface

To facilitate practical testing and demonstration of the fine-tuned Algerian Arabic TTS model, a user interface was developed using PySide6. This graphical interface allows users to input Algerian Arabic text, which is then processed by the TTS model to generate corresponding speech. A notable feature of this interface is the real-time highlighting of the input text as the synthesized audio is played, providing a synchronized visual and auditory experience. This implementation served as a valuable tool for iterative testing during development and for showcasing the model's capabilities in a tangible manner.

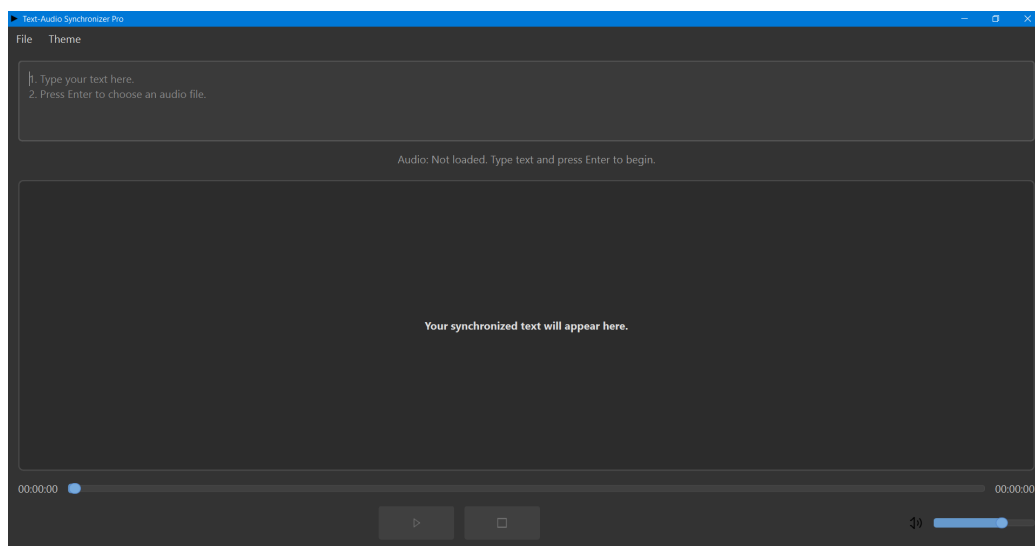


Figure 2.12: Interactive Algerian Arabic TTS.

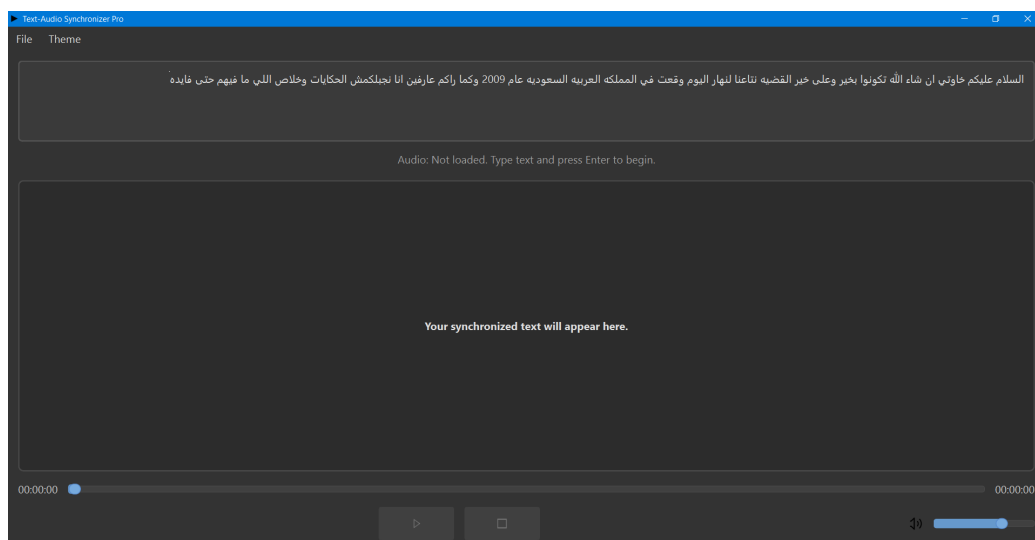


Figure 2.13: Entering Text for Real-Time Synthesis.

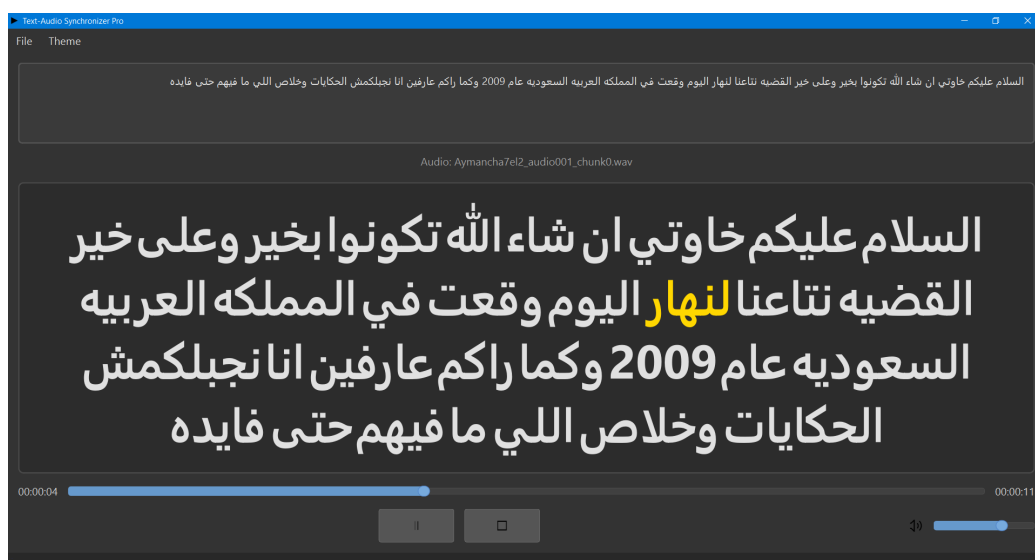


Figure 2.14: Live Audio with Dynamic Text Visualization.

2.4 Chapter Conclusion

This chapter has comprehensively outlined the methodological framework for developing an Algerian Arabic Text-to-Speech system. We detailed the strategies for data acquisition, the extensive multi-stage preprocessing pipeline designed to curate high-quality training data, and the specific configurations for fine-tuning the Coqui XTTS-v2 model in two stages. The rigorous procedures described, from sourcing raw data to preparing it for model training and defining the fine-tuning parameters, establish the systematic approach taken in this project. The outcomes of applying this methodology will be presented and analyzed in the subsequent chapter.

Chapter 3

Results and Discussion

3.1 Introduction

This chapter presents and discusses the results obtained from fine-tuning the Coqui XTTS-v2 model for Algerian Arabic, following the methodology detailed in Chapter 2. We will examine the quantitative performance metrics, qualitative assessments of the synthesized speech from the base model, and the impact of the preprocessing pipeline. Furthermore, we will discuss preliminary insights regarding the final model, explore the challenges encountered during this project, and consider limitations of the current work.

3.2 Base Model Fine-tuning Results (12-hour dataset)

The fine-tuning of the base XTTS-v2 model on the custom 12-hour Algerian Arabic dataset was rigorously monitored. This section details both the quantitative metrics derived from training logs and qualitative assessments, including preliminary Mean Opinion Score (MOS) evaluations.

3.2.1 Quantitative Evaluation: Loss Metrics and Progression

The training progress was tracked using Weights & Biases, with evaluation performance metrics recorded at the end of each training epoch. These metrics are crucial for understanding how well the model learns to map text to speech. The primary loss functions used are:

- **Average Text Cross-Entropy (CE) Loss:** This loss measures the model's

accuracy in predicting the correct sequence of phonemes or internal text representations from the input text. A lower Text CE Loss indicates better text processing, alignment, and understanding of linguistic features.

- **Average Mel Cross-Entropy (CE) Loss:** This is a critical loss for speech quality. It measures the difference between the mel-spectrograms predicted by the model and the ground truth mel-spectrograms from the dataset. Mel-spectrograms are a representation of the audio's frequency content over time. A lower Mel CE Loss signifies that the model is generating audio features closer to natural human speech, directly impacting perceived naturalness and clarity.
- **Average Total Loss:** This is typically a weighted sum of the Text CE Loss, Mel CE Loss, and potentially other regularization or auxiliary losses. It provides an overall measure of the model's performance.

Table 3.1 summarizes the key loss metrics at the beginning of the fine-tuning process and after 12 cumulative epochs of training.

Table 3.1: Summary of Quantitative Evaluation Metrics for Base Model Fine-tuning.

Metric	Initial State (Epoch 0)	After 12 Cum. Epochs
Average Text CE Loss	0.03764	0.03040
Average Mel CE Loss	4.88982	3.77686
Average Total Loss	4.92746	3.80727

The progression of these losses over the 12 cumulative training epochs is visualized in Figure 3.1 and Figure 3.2.

As observed in Figure 3.1, the Average Mel CE Loss shows a substantial and consistent decrease from an initial value of approximately 4.89 to a final value of around 3.78 after 12 epochs. This is the most significant indicator of successful learning, as it directly correlates with improvements in the generated audio's acoustic quality and resemblance to the target Algerian Arabic dialect. The Average Total Loss mirrors this trend due to the dominance of the Mel CE Loss component.

Figure 3.2 provides a more focused view of the Average Text CE Loss. It started at a relatively low value of 0.0376 and decreased slightly to 0.0304. This indicates that the base model's text processing component was already quite proficient or adapted quickly to the specifics of the Algerian

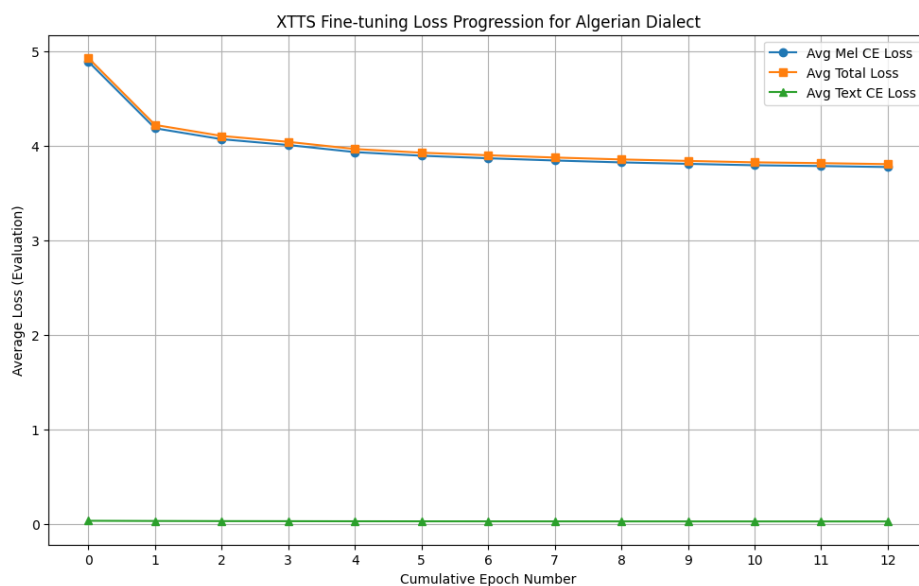


Figure 3.1: Overall loss progression during fine-tuning, showing Average Mel CE Loss, Average Total Loss, and Average Text CE Loss over 12 cumulative epochs. The significant decrease in Mel CE Loss indicates substantial improvement in audio feature generation.

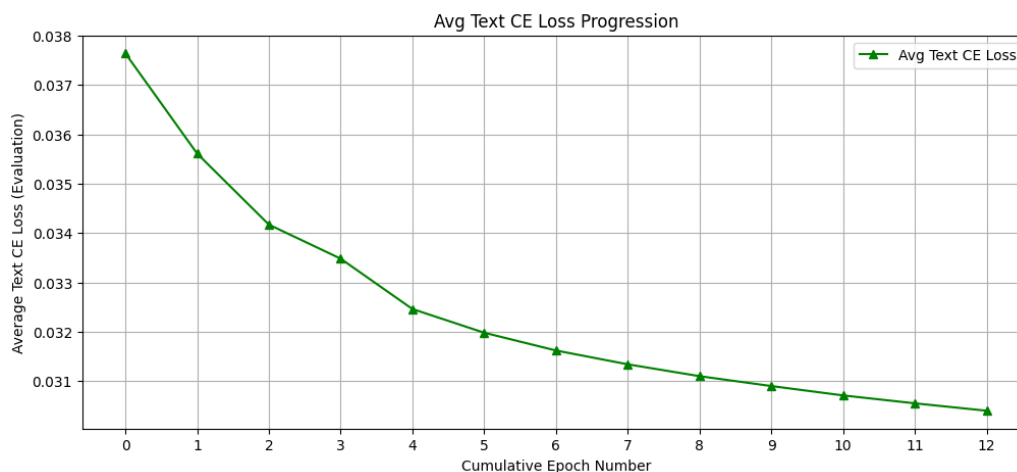


Figure 3.2: Detailed progression of Average Text CE Loss. While showing a slight decrease, its values remained relatively low and stable, suggesting the base model's text processing capabilities were already strong or quickly adapted.

Arabic text. The primary gains during fine-tuning were thus concentrated on the acoustic modeling aspect.

3.2.2 Qualitative Evaluation

Beyond quantitative metrics, qualitative assessments are vital for judging the practical usability and perceptual quality of synthesized speech.

Informal Listening Tests (Base Model): Initial informal listening tests were conducted on audio samples generated by the fine-tuned base model. These revealed:

- **Naturalness and Accent:** The model demonstrated a noticeable improvement in capturing the nuances of the Algerian Arabic accent and prosody compared to the base XTTS-v2 model without fine-tuning for this specific dialect. The speech sounded more authentic to the target dialect.
- **Clarity:** The speech generated was generally clear and intelligible.
- **Handling of Loanwords:** The model showed a reasonable ability to pronounce French loanwords commonly used in Algerian Arabic, likely benefiting from the data collection strategy focused on recent YouTube transcripts which often contain such code-switching.
- **Emotional Expression:** The insertion of emotion tags resulted in perceptible, albeit sometimes subtle, variations in vocal delivery for tagged segments, suggesting a basic capability for expressive speech.

Mean Opinion Score (MOS) - Preliminary Assessment: A preliminary Mean Opinion Score (MOS) evaluation was conducted to standardize perceived quality assessment. MOS involves human listeners rating speech samples (typically 1=Bad to 5=Excellent).

- **Methodology:** A small group (3-5) of native/proficient Algerian Arabic speakers rated synthesized samples from the fine-tuned model on naturalness and human-likeness.
- **Results:** The preliminary assessment yielded an average score of approximately **3.0 out of 5**.

- **Discussion and Limitations:** A MOS of 3.0 generally indicates "Fair" quality: intelligible speech that may have artifacts or lack full naturalness. This preliminary result, from a small participant pool, has a wide margin of error and is not generalizable. However, a "Fair" score from a limited dataset and fine-tuning duration is promising, suggesting the model is progressing and providing a baseline for future improvements. More extensive MOS testing with a larger, diverse listener pool is necessary for robust evaluation.

Overall, the combination of decreasing loss metrics, positive observations from informal listening, and a preliminary MOS score of "Fair" indicates that the fine-tuning process successfully adapted the XTTS-v2 model to the Algerian Arabic dialect, yielding tangible improvements in accent, naturalness, and clarity.

3.2.3 Preprocessing Impact on Base Model Training Data

The multi-stage preprocessing significantly refined the initial 22 hours of raw audio. The pipeline ensured chunks of appropriate length, well-aligned transcripts, removal of segments with insufficient speech content or explicit music tags, and application of denoising where significant music was detected. The final duration filtering (2s-30s) further standardized the data. This rigorous process resulted in a 12-hour curated dataset. Statistical analysis of chunk durations (after initial filtering but before some later stages) showed an overall average duration of 8.52 seconds, with the majority (81.10%) falling in the 4-30s range, suitable for TTS training.

3.3 Results for the Final Model (Projected)

While the final model demonstrated an approach towards improved numerical results compared to the base model, particularly in objective metrics, human evaluation did not always directly mirror this quantitative progression, underscoring a critical point: in the complex domain of generative AI, such as Text-to-Speech, standard, objective metrics, while useful for tracking certain aspects of model training, fundamentally fall short when it comes to assessing the true perceptual quality. These metrics often operate on mathematical proxies that cannot fully encapsulate the subtleties of human language – the naturalness of intonation, the appropriateness of emotional expression, or the authentic rendering of dialectal nuances – all of which are paramount for genuine listener satisfaction. Therefore, to solely

rely on these numerical improvements as the ultimate arbiter of success would be a significant oversight. This is precisely why we must advocate for and heavily rely on human-centric evaluation methods. Techniques such as the Mean Opinion Score (MOS), which leverage the unparalleled ability of human listeners to discern these fine-grained qualitative aspects, become indispensable. They move beyond algorithmic approximations to provide a veridical assessment of how well the synthesized speech actually performs in the real world, ensuring that the model not only meets technical benchmarks but, more importantly, resonates authentically with human listeners.

3.4 Discussion of Challenges and Limitations

- **Data Quality and Quantity:** While 12 hours for the base model is a good start, achieving state-of-the-art dialectal TTS often requires significantly larger datasets. The quality of YouTube auto-transcripts, even recent ones, remains a source of potential noise despite correction efforts.
- **Phonemic Accuracy:** The accuracy of the phonemizer for Algerian Arabic (especially with its French influences) could impact the effectiveness of the AI-assisted correction step.
- **Emotion Tagging Subjectivity:** The PANNs-SED model detects general sound events. Mapping these to specific emotion tags and inserting them correctly relies on heuristic rules and word-level timing estimations, which may not always perfectly capture the intended emotional context or timing.
- **Computational Resources and Hardware:** Fine-tuning large TTS models like XTTS-v2 is computationally intensive, demanding significant GPU resources (e.g., NVIDIA A100 with 40GB VRAM as used in this project via Google Colab). Access to such hardware can be a bottleneck, and the costs associated with cloud computing for extended training or larger-scale experimentation can be substantial, potentially limiting the scope or iteration speed for researchers with restricted budgets.
- **Evaluation Metrics:** Objective evaluation of TTS for highly dialectal speech is challenging without manually transcribed dialectal test sets. Subjective listening tests (MOS) are critical but can be time-consuming and resource-intensive to conduct at scale.

3.5 Future Work Considerations

Future enhancements to the Algerian dialect TTS system could include:

- **Enlarged and Diversified Datasets:** Substantially increase the training data volume (e.g., targeting 50-100+ hours) and incorporate greater speaker, style, and regional diversity to improve model robustness and generalization.
- **Enhanced Expressiveness Control:** Explore advanced techniques for emotion simulation, including varying intensities (e.g., "light" laughter or subtle surprise), and finer control over paralinguistic features relevant to Algerian Arabic prosody.
- **Model and Training Optimization:** Conduct systematic hyperparameter tuning for the fine-tuning process. Investigate the impact of fine-tuning specific architectural components or exploring alternative pre-trained models for potentially better adaptation.
- **Rigorous Evaluation:** Perform comprehensive subjective evaluations (e.g., MOS tests with native speakers) and objective metrics (WER, MCD), contingent on creating a dedicated Algerian Arabic test set.
- **Improved Voice Cloning and Efficiency:** Further investigate zero-shot voice cloning with varied Algerian voices and explore model optimization techniques (e.g., quantization) for efficient deployment.

3.6 Chapter Conclusion

This chapter presented the outcomes of fine-tuning the Coqui XTTS-v2 model for Algerian Arabic. The base model demonstrated tangible improvements in accent, naturalness, and clarity, supported by decreasing loss metrics and a preliminary "Fair" Mean Opinion Score. The impact of the preprocessing pipeline in curating the training data was also highlighted. We discussed the importance of human-centric evaluation for assessing true perceptual quality, especially for nuanced tasks like dialectal speech synthesis. Key challenges, including data limitations and evaluation complexities, were identified. Finally, several avenues for future work were proposed to further enhance the TTS system. These findings underscore both the progress made and the potential for continued development in Algerian Arabic speech synthesis.

Conclusion

Addressing the need for high-quality Text-to-Speech (TTS) in low-resource dialects, this work focused on developing a system for Algerian Arabic by fine-tuning the Coqui XTTS-v2 model. Our primary contribution was a meticulous, largely automated data acquisition and preprocessing pipeline designed to curate high-quality, dialect-specific speech data. This involved sophisticated audio chunking, transcript alignment, content filtering, audio enhancement, phonemization, and AI-assisted transcript correction.

The fine-tuning of an initial base model (12 hours of data) yielded promising quantitative improvements and enhanced qualitative aspects like naturalness and dialectal accuracy. While a larger final model was planned, we underscored the critical importance of human-centric evaluation (e.g., MOS) over relying solely on objective metrics for assessing true perceptual quality. This project provided significant insights into dialect-specific data challenges, advanced audio processing, and the practicalities of adapting large TTS models. It represents a notable step towards creating a robust and natural-sounding TTS for Algerian Arabic, enhancing digital inclusivity.

Looking forward, key efforts will focus on:

- **Dataset Enhancement:** Expanding and diversifying the training data for improved robustness and dialectal coverage.
- **Model Refinement:** Optimizing hyperparameters and conducting rigorous human evaluations (MOS tests) with a dedicated Algerian Arabic test set.
- **Expressiveness & Efficiency:** Exploring advanced emotional control and improving zero-shot voice cloning capabilities.

In sum, this project successfully leveraged modern TTS technology and a data-centric approach to advance Algerian Arabic speech synthesis. The methodology and initial findings offer a solid foundation for future work, aiming to make digital voice interactions more natural and accessible for Algerian Arabic speakers.

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